

Solo E-Commerce Platform - Complete Project Documentation

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1. Project Overview

1.1 Description

Solo is a full-featured B2C e-commerce platform designed for a premium home and lifestyle products retail business. The platform provides a complete online shopping experience with a modern storefront, comprehensive admin dashboard, content management system (CMS), and inventory management.

1.2 Scope

- **Customer Portal:** Product browsing, search, cart, checkout, order management, account management
- **Admin Dashboard:** Product management, order management, customer management, CMS, analytics
- **CMS:** Landing pages, banners, blog, navigation menus, homepage configuration
- **Loyalty Program:** Customer rewards system with cashback
- **Multi-Schema Database:** Separation of concerns between public app data and inventory catalog

1.3 Repository

- **GitHub:** <https://github.com/anydevice1234123-netizen/Solo-Jan-13>
- **Current Branch:** main

2. Technology Stack

2.1 Backend

Technology	Purpose	Version
NestJS	Backend Framework	^10.x
TypeScript	Programming Language	^5.x
Prisma	ORM (Object-Relational Mapping)	^5.x
PostgreSQL	Database	15+
JWT	Authentication	-
Argon2id	Password Hashing	-
@nestjs/throttler	Rate Limiting	-
Sharp	Image Processing	-
PDFKit	Invoice Generation	-

2.2 Frontend

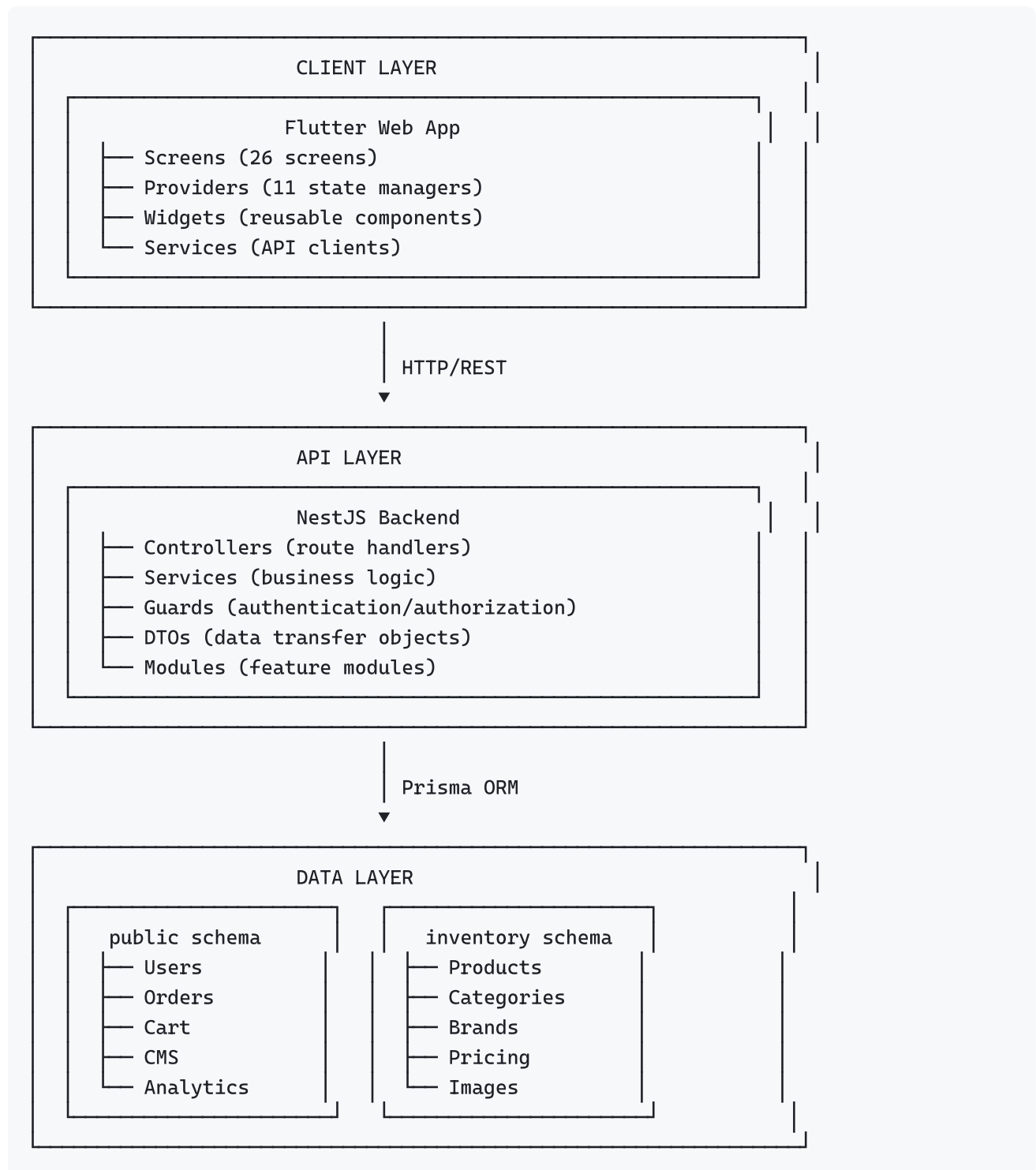
Technology	Purpose	Version
Flutter	UI Framework	^3.x
Dart	Programming Language	^3.x
Provider	State Management	-
HTTP	API Communication	-
url_launcher	External Links	-
flutter_svg	SVG Support	-

2.3 Infrastructure

Technology	Purpose
Node.js	Runtime Environment
npm/yarn	Package Management
Local File System	Media Storage

3. Architecture Overview

3.1 System Architecture



3.2 Backend Module Structure

```
backend/src/
├── admin/           # Admin dashboard APIs
├── analytics/       # Analytics tracking
├── auth/            # Authentication & authorization
├── blog/            # Blog management
├── brands/          # Brand management
├── cart/            # Shopping cart
├── catalog/         # Product catalog (inventory integration)
├── categories/      # Category management
├── cms/             # CMS home/category landing pages
├── collections/     # Product collections
├── common/          # Shared utilities, guards, decorators
├── content/         # Banners, landing pages
├── customers/       # Customer management
├── debug/           # Debug utilities
├── departments/     # Department management
├── email/           # Email services
├── media/           # File upload & management
├── navigation/      # Navigation menus
├── orders/          # Order processing & invoices
├── packages/        # Product bundles
├── prisma/          # Prisma service
├── products/        # Product management
├── promos/          # Promo codes
└── users/           # User management
```

3.3 Frontend Structure

```
frontend/lib/
├── config/           # App configuration
├── core/             # Core utilities
│   ├── api/         # API clients
│   ├── cache/       # Caching
│   ├── dto/         # Data transfer objects
│   └── events/      # Event bus
├── guards/          # Route guards
├── layouts/         # Layout components
├── models/          # Data models
│   └── dto/         # DTO models
├── providers/       # State management (11 providers)
├── screens/         # UI screens (26 screens)
│   └── admin/       # Admin screens (16 screens)
├── services/        # Business services
├── theme/           # App theming
└── widgets/        # Reusable widgets
    ├── admin/      # Admin widgets
    ├── app_drawer/ # Drawer components
    ├── app_header/ # Header components
    ├── cms/        # CMS widgets
    ├── home/       # Home page widgets
    └── porto/      # Porto-style widgets
```

4. Database Schema & ER Diagram

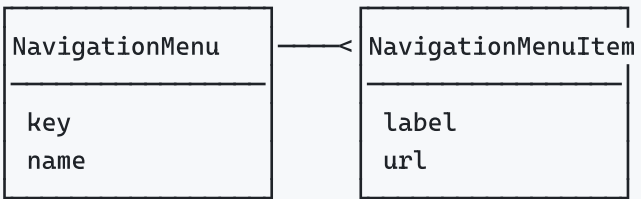
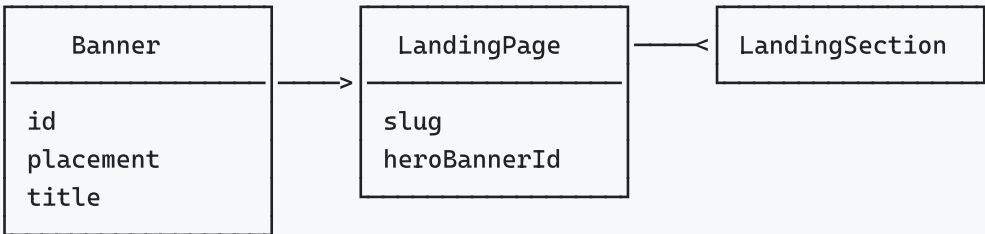
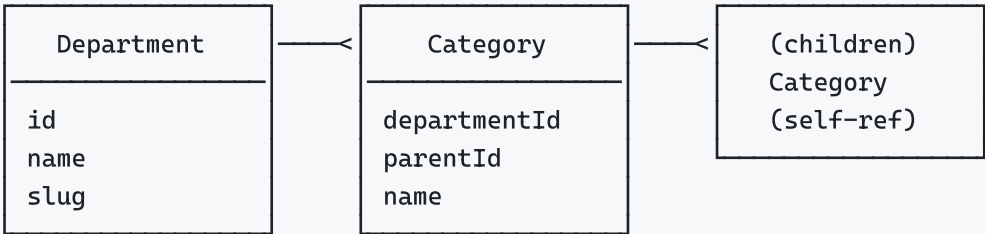
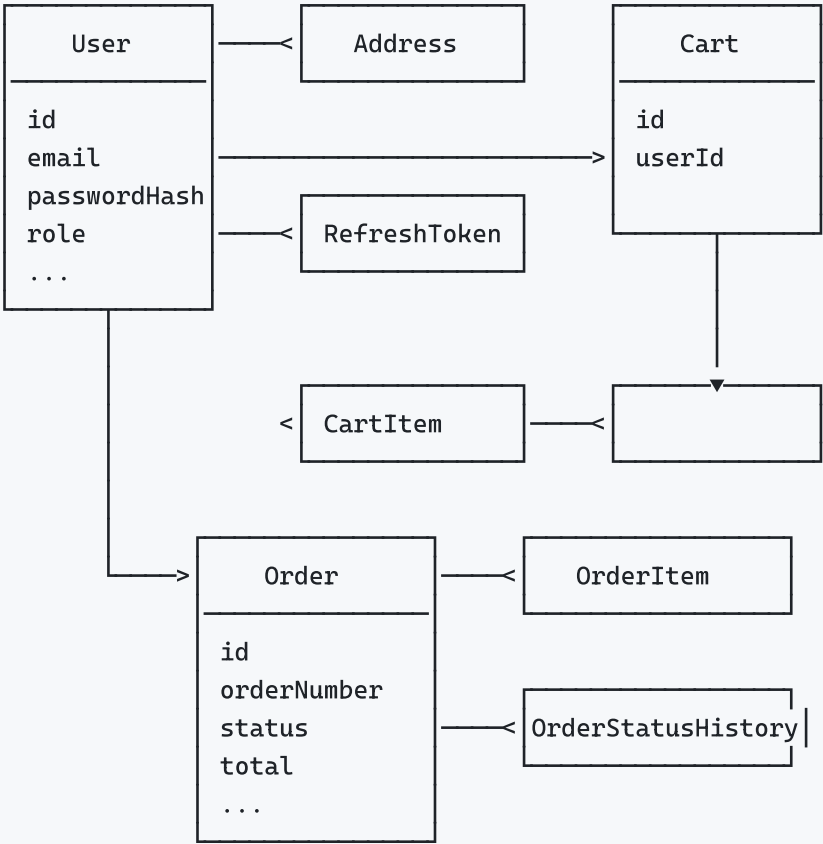
4.1 Schema Overview

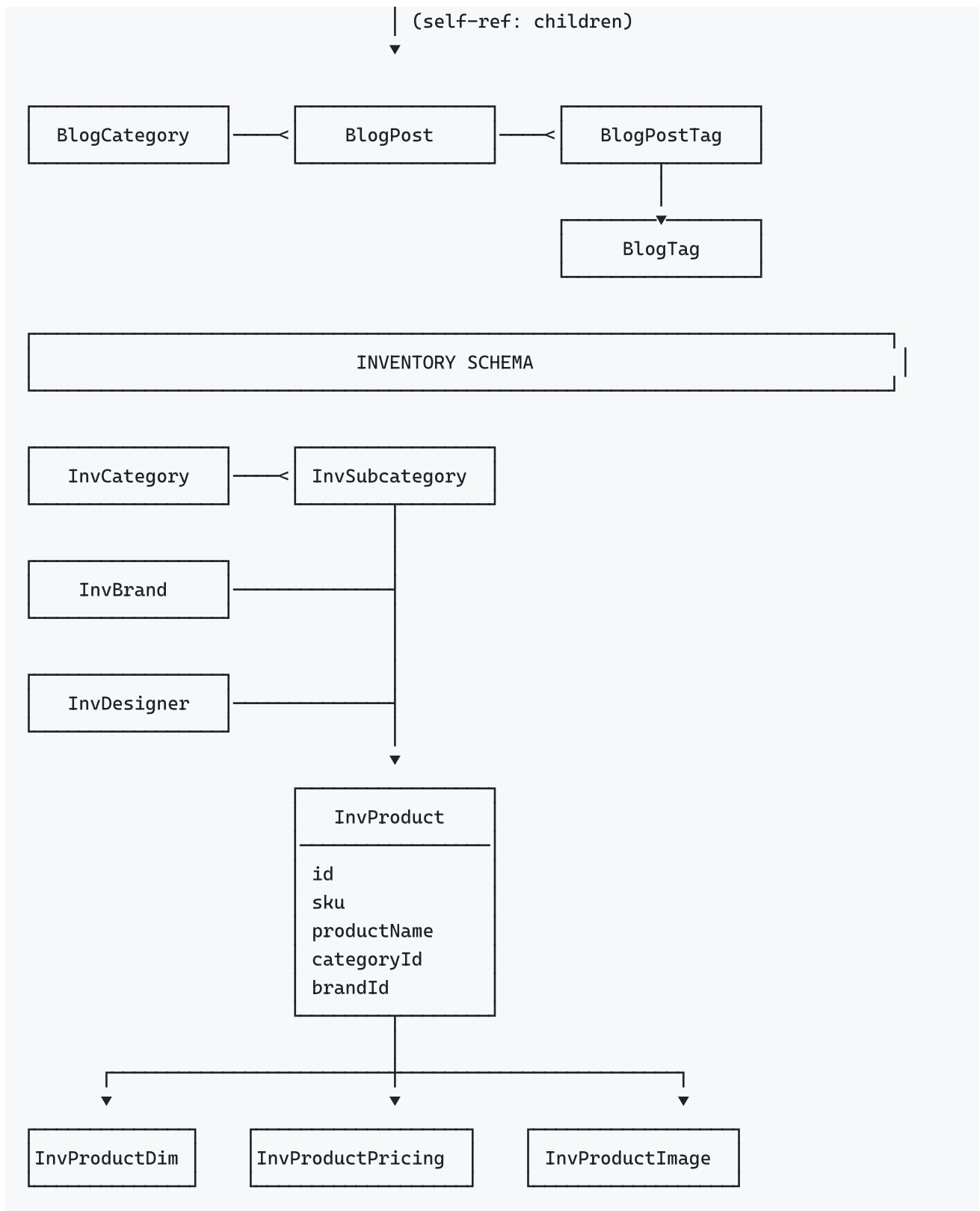
The database uses PostgreSQL with Prisma's multi-schema feature, separating data into two schemas:

1. **public schema:** Application data (users, orders, CMS, analytics)
2. **inventory schema:** Product catalog data (products, pricing, inventory)

4.2 Entity Relationship Diagram (Text Representation)

PUBLIC SCHEMA





4.3 Model Count Summary

Schema	Model Count	Key Models
public	~35 models	User, Order, Cart, Category, Brand, Banner, LandingPage, BlogPost, NavigationMenu, PromoCode, AnalyticsEvent
inventory	~10 models	InvProduct, InvCategory, InvSubcategory, InvBrand, InvProductPricing, InvProductImage

4.4 Key Relationships

1. **User** → **Order**: One-to-Many (user can have multiple orders)
2. **User** → **Cart**: One-to-One (user has one active cart)
3. **User** → **Address**: One-to-Many (user can have multiple addresses)
4. **User** → **LoyaltyWallet**: One-to-One (user has one loyalty wallet)
5. **Department** → **Category**: One-to-Many
6. **Category** → **Category**: Self-referential (parent-child hierarchy)
7. **InvCategory** → **InvSubcategory**: One-to-Many
8. **InvProduct** → **InvProductPricing**: One-to-One
9. **InvProduct** → **InvProductImage**: One-to-Many
10. **Order** → **OrderItem**: One-to-Many
11. **LandingPage** → **LandingSection**: One-to-Many
12. **NavigationMenu** → **NavigationMenuItem**: One-to-Many (hierarchical)

5. Backend API Documentation

5.1 Authentication APIs (/auth)

Method	Endpoint	Description	Auth
POST	/auth/register	Register new user	No
POST	/auth/login	Login with email/password	No
POST	/auth/refresh	Refresh access token	No
POST	/auth/logout	Logout (revoke refresh token)	Yes
GET	/auth/me	Get current user profile	Yes
POST	/auth/change-password	Change password	Yes
POST	/auth/forgot-password	Request password reset	No
POST	/auth/reset-password	Reset password with token	No
POST	/auth/verify-email	Verify email address	No

Method	Endpoint	Description	Auth
POST	/auth/resend-verification	Resend verification email	No

5.2 Products APIs (/products)

Method	Endpoint	Description	Auth
GET	/products	List all products with filters	No
GET	/products/featured	Get featured products	No
GET	/products/best-sellers	Get best-selling products	No
GET	/products/new-arrivals	Get new arrivals	No
GET	/products/:id/related	Get related products	No
GET	/products/:slugOrId	Get single product	No
GET	/products/inventory/categories	Get inventory categories	No
GET	/products/inventory/brands	Get inventory brands	No
POST	/products	Create product override	Admin
PATCH	/products/:id	Update product override	Admin
DELETE	/products/:id	Delete product override	Admin

5.3 Categories APIs (/categories)

Method	Endpoint	Description	Auth
GET	/categories	List all categories	No
GET	/categories/:id	Get single category	No
POST	/categories	Create category	Admin
PATCH	/categories/:id	Update category	Admin
DELETE	/categories/:id	Delete category	Admin

5.4 Cart APIs (/cart)

Method	Endpoint	Description	Auth
GET	/cart	Get user's cart	Yes
POST	/cart/items	Add item to cart	Yes

Method	Endpoint	Description	Auth
PATCH	/cart/items/:id	Update cart item quantity	Yes
DELETE	/cart/items/:id	Remove item from cart	Yes
DELETE	/cart	Clear entire cart	Yes

5.5 Orders APIs (/orders)

Method	Endpoint	Description	Auth
POST	/orders	Create new order	Yes
GET	/orders	Get user's orders	Yes
GET	/orders/:id	Get order details	Yes
GET	/orders/:id/invoice/pdf	Download invoice PDF	Yes

5.6 Content/CMS APIs (/content)

Method	Endpoint	Description	Auth
GET	/content/home	Get homepage layout	No
GET	/content/banners	Get active banners	No
GET	/content/pages/:slug	Get landing page by slug	No
GET	/content/loyalty-config	Get loyalty page config	No
GET	/content/admin/banners	List all banners	Admin
POST	/content/admin/banners	Create banner	Admin
PATCH	/content/admin/banners/:id	Update banner	Admin
DELETE	/content/admin/banners/:id	Delete banner	Admin
GET	/content/admin/pages	List all landing pages	Admin
POST	/content/admin/pages	Create landing page	Admin
PATCH	/content/admin/pages/:id	Update landing page	Admin
DELETE	/content/admin/pages/:id	Delete landing page	Admin

5.7 CMS Home/Category APIs (/cms)

Method	Endpoint	Description	Auth
GET	/cms/home-page	Get home page CMS	No
GET	/cms/category/:categoryId	Get category landing	No
GET	/cms/admin/home	Get home page admin	Admin
POST	/cms/admin/home/sections	Create home section	Admin
PATCH	/cms/admin/home/sections/:id	Update home section	Admin
DELETE	/cms/admin/home/sections/:id	Delete home section	Admin
POST	/cms/admin/home/sections/reorder	Reorder sections	Admin

5.8 Blog APIs (/blog)

Method	Endpoint	Description	Auth
GET	/blog	List blog posts	No
GET	/blog/sidebar	Get sidebar data	No
GET	/blog/recent	Get recent posts	No
GET	/blog/post/:slug	Get post by slug	No
GET	/blog/categories	List categories	No
GET	/blog/tags	List tags	No
GET	/blog/admin/posts	Admin list posts	Admin
POST	/blog/admin/posts	Create post	Admin
PATCH	/blog/admin/posts/:id	Update post	Admin
DELETE	/blog/admin/posts/:id	Delete post	Admin

5.9 Navigation APIs (/navigation)

Method	Endpoint	Description	Auth
GET	/navigation/menu/:key	Get menu by key	No
GET	/navigation/admin/menus	List all menus	Admin
POST	/navigation/admin/menus	Create menu	Admin
PATCH	/navigation/admin/menus/:id	Update menu	Admin
DELETE	/navigation/admin/menus/:id	Delete menu	Admin

Method	Endpoint	Description	Auth
POST	/navigation/admin/items	Create menu item	Admin
PATCH	/navigation/admin/items/:id	Update menu item	Admin
DELETE	/navigation/admin/items/:id	Delete menu item	Admin

5.10 Admin APIs (/admin)

Method	Endpoint	Description	Auth
GET	/admin/stats	Get dashboard stats	Admin
GET	/admin/orders	List all orders	Admin
GET	/admin/orders/:id	Get order details	Admin
GET	/admin/orders/:id/invoice/pdf	Download invoice	Admin

5.11 Media APIs (/media)

Method	Endpoint	Description	Auth
POST	/media/upload	Upload single file	Admin
POST	/media/upload-multiple	Upload multiple files	Admin
DELETE	/media/delete	Delete file	Admin

6. Frontend Documentation

6.1 Screen Inventory (26 Screens)

Customer-Facing Screens

Screen	File	Description
Home	home_screen.dart	Main landing page with CMS sections
Product List	product_list_screen.dart	Product listing with filters
Product Detail	product_detail_screen.dart	Individual product view
Category	category_screen.dart	Category browse page
Category Landing	category_landing_screen.dart	Category CMS page
Search	search_screen.dart	Product search

Screen	File	Description
Cart	<code>cart_screen.dart</code>	Shopping cart
Checkout	<code>checkout_screen.dart</code>	Checkout flow
Order Confirmation	<code>order_confirmation_screen.dart</code>	Post-order success
Favorites	<code>favorites_screen.dart</code>	Wishlist
Login	<code>login_screen.dart</code>	User login
Signup	<code>signup_screen.dart</code>	User registration
Account Shell	<code>account/account_shell.dart</code>	Account management
About Us	<code>about_us_screen.dart</code>	About page
Loyalty Program	<code>loyalty_program_screen.dart</code>	Rewards program
Blog	<code>blog_screen.dart</code>	Blog listing
Blog Post	<code>blog_post_screen.dart</code>	Individual blog post
Bulk Order	<code>bulk_order_screen.dart</code>	Bulk ordering

Admin Screens (16 Screens)

Screen	File	Description
Admin Dashboard	<code>admin_dashboard_screen.dart</code>	Dashboard with stats
Admin Login	<code>admin_login_screen.dart</code>	Admin authentication
Products Management	<code>admin_products_screen.dart</code>	Product listing
Product Form	<code>admin_product_form_screen.dart</code>	Create/edit product
Categories Management	<code>admin_categories_screen.dart</code>	Category listing
Category Form	<code>admin_category_form_screen.dart</code>	Create/edit category
Brands Management	<code>admin_brands_screen.dart</code>	Brand listing
Departments	<code>admin_departments_screen.dart</code>	Department management
Orders Management	<code>admin_orders_screen.dart</code>	Order listing
Order Details	<code>admin_order_details_screen.dart</code>	Order view
Customers	<code>admin_customers_screen.dart</code>	Customer listing
Customer Details	<code>admin_customer_details_screen.dart</code>	Customer view
Banners	<code>admin_banners_screen.dart</code>	Banner management

Screen	File	Description
Landing Pages	<code>admin_landing_pages_screen.dart</code>	CMS pages
Section Editor	<code>section_editor_shell.dart</code>	Section editing
Generic List	<code>admin_generic_list_screen.dart</code>	Reusable list

6.2 State Management (Providers)

Provider	File	Purpose
AuthProvider	<code>auth_provider.dart</code>	Authentication state
AccountProvider	<code>account_provider.dart</code>	User account data
CartProvider	<code>cart_provider.dart</code>	Shopping cart state
CatalogProvider	<code>catalog_provider.dart</code>	Categories & brands
ContentProvider	<code>content_provider.dart</code>	CMS content
FavoritesProvider	<code>favorites_provider.dart</code>	Wishlist
HomeProvider	<code>home_provider.dart</code>	Home page sections
HomeCmsProvider	<code>home_cms_provider.dart</code>	CMS home data
ProductDetailsProvider	<code>product_details_provider.dart</code>	Product details
ProductListProvider	<code>product_list_provider.dart</code>	Product listing
SearchProvider	<code>search_provider.dart</code>	Search functionality

6.3 API Client Architecture

```
// Centralized API Service Factory
class Api {
  static ApiClient client;

  // API instances
  static ProductsApi products;
  static CategoriesApi categories;
  static BrandsApi brands;
  static DepartmentsApi departments;
  static ContentApi content;
  static MediaApi media;
  static AuthApi auth;
  static AdminApi admin;
}
```

6.4 Key Widgets

Widget	Purpose
ModernDrawer	Navigation drawer
AppHeader	App header with cart/search
TopBanner	Promotional banner
HeroBanner	Hero carousel
ProductCard	Product display card
PortoProductCard	Porto-style product card
CmsSectionsBuilder	Dynamic CMS renderer
LoyaltyProgramBanner	Loyalty promotion
AdminLayout	Admin panel layout

7. Features & Functionalities

7.1 Customer Features

Shopping Experience

- **Product Browsing:** Browse by category, brand, or collection
- **Product Search:** Full-text search with filters
- **Product Details:** Images, descriptions, specs, pricing
- **Related Products:** Cross-selling suggestions
- **Wishlist/Favorites:** Save products for later

Cart & Checkout

- **Shopping Cart:** Add, update, remove items
- **Quantity Management:** Adjust quantities
- **Cart Persistence:** Synced with server for logged-in users
- **Checkout Flow:** Multi-step checkout
- **Address Management:** Save multiple addresses
- **Payment Methods:** Credit card, COD
- **Promo Codes:** Discount application
- **Loyalty Cash:** Redeem rewards at checkout

Account Management

- **Registration/Login:** Email-based authentication
- **Password Management:** Change, reset, forgot
- **Email Verification:** Verify email address
- **Profile Management:** Update personal info
- **Address Book:** Manage addresses
- **Order History:** View past orders
- **Invoice Download:** PDF invoices

Loyalty Program

- **Cashback Rewards:** Earn % on purchases
- **Wallet Balance:** View & track rewards
- **Redemption:** Apply at checkout
- **Transaction History:** View earning/spending

7.2 Admin Features

Dashboard

- **Stats Overview:** Revenue, orders, customers
- **Recent Orders:** Quick order view
- **Real-time Clock:** Current time display
- **Quick Actions:** Common admin tasks

Product Management

- **Product Listing:** Paginated product list
- **Search & Filter:** Find products
- **Product Override:** Customize inventory products
- **Featured/New/Best Seller:** Merchandising flags
- **Custom Pricing:** Override inventory prices
- **Image Management:** Upload product images

Order Management

- **Order Listing:** All orders with status
- **Order Details:** Full order view
- **Status Updates:** Change order status
- **Invoice Generation:** PDF invoices

Customer Management

- **Customer List:** All registered users
- **Customer Details:** Profile & order history
- **Address View:** Customer addresses

CMS Management

- **Banners:** Hero, promo, category banners
- **Landing Pages:** Custom page builder
- **Page Sections:** Flexible content blocks
- **Navigation Menus:** Header/footer menus
- **Blog:** Posts, categories, tags

Category/Brand Management

- **Categories:** Hierarchical categories
- **Brands:** Brand listing & logos
- **Departments:** Top-level organization

7.3 Content Management

Banner Types

- `HOME_HERO` - Main hero slider
- `HOME_SECONDARY` - Secondary home banners
- `HOME_MID` - Mid-page promotions
- `CATEGORY_TOP` - Category header
- `PROMOTION` - Promotional banners

Landing Page Section Types

- `HERO` / `HERO_SLIDER` - Hero area
- `CATEGORY_TILES` - Category grid
- `PRODUCT_CAROUSEL` - Product slider
- `BRAND_STRIP` - Brand logos
- `PROMO_BANNER` - Promotional
- `PRODUCT_GRID` - Product grid
- `RICH_TEXT` - HTML content
- `TESTIMONIALS` - Reviews
- `NEWSLETTER_BLOCK` - Email signup

8. Security Implementation

8.1 Authentication

- **JWT Tokens:** Access & refresh token pair
- **Token Refresh:** Automatic token renewal
- **Password Hashing:** Argon2id algorithm

- **Email Verification:** Token-based verification

8.2 Authorization

- **Role-Based Access:** CUSTOMER, ADMIN, SUPER_ADMIN
- **Route Guards:** JWT validation
- **Role Guards:** Permission checking
- **Admin Routes:** Protected endpoints

8.3 Rate Limiting

- **Global Throttling:** 1000 requests/minute
- **Login Throttling:** 5 attempts/15 minutes
- **Registration Throttling:** 3 attempts/hour

8.4 Input Validation

- **DTOs:** Class-validator decorators
- **File Upload:** Type & size validation
- **SQL Injection:** Prisma ORM protection
- **XSS Prevention:** Input sanitization

9. Deployment & Configuration

9.1 Environment Variables

```
# Database
DATABASE_URL=postgresql://user:pass@host:5432/db?schema=public

# JWT
JWT_SECRET=your-secret-key
JWT_EXPIRES_IN=15m
REFRESH_TOKEN_EXPIRES_IN=7d

# App
PORT=3000
NODE_ENV=development

# Media
UPLOAD_PATH=./uploads
MAX_FILE_SIZE=10485760
```

9.2 Running the Application

Backend:

```
cd backend
npm install
npx prisma generate
npx prisma migrate deploy
npx prisma db seed
npm run start:dev
```

Frontend:

```
cd frontend
flutter pub get
flutter run -d chrome --web-port=5000
```

9.3 Database Migrations

```
# Generate migration
npx prisma migrate dev --name migration_name

# Apply migrations
npx prisma migrate deploy

# Seed database
npx prisma db seed
```

9.4 Ports

- **Backend API:** <http://localhost:3000>
- **Frontend Web:** <http://localhost:5000>

Appendix A: Enum Reference

User Roles

- `CUSTOMER` - Regular customer
- `ADMIN` - Admin access
- `SUPER_ADMIN` - Full access

Order Status

- `PENDING` - Order created
- `PAYMENT_PENDING` - Awaiting payment

- `PAID` - Payment received
- `PROCESSING` - Being processed
- `SHIPPED` - In transit
- `DELIVERED` - Delivered
- `CANCELLED` - Cancelled
- `REFUNDED` - Refunded

Payment Status

- `PENDING` - Awaiting payment
- `AUTHORIZED` - Payment authorized
- `PAID` - Payment complete
- `FAILED` - Payment failed
- `REFUNDED` - Payment refunded

Payment Methods

- `CREDIT_CARD` - Card payment
- `CASH_ON_DELIVERY` - COD

Shipping Methods

- `STANDARD` - Standard delivery
- `EXPRESS` - Express delivery
- `OVERNIGHT` - Next-day delivery
- `PICKUP` - Store pickup

Appendix B: Model Count

Category	Count
Backend Modules	20+
API Endpoints	80+
Database Models	45+
Frontend Screens	26
Frontend Providers	11
Admin Screens	16

Documentation generated for Solo E-Commerce Platform Last Updated: January 2025